

Practical 9

Data Warehousing and OLAP Cube Development

Part 1: Create Data Warehouse (SSMS)

Step 1: Open SSMS

- Open SQL Server Management Studio and connect to Database Engine.

Step 2: New Query

- Click New Query to open the query editor.

Step 3: Run SQL Scripts

- Paste the given SQL scripts one by one and press F5 to run each.

Step 4: Result

- Database 'Sales_DW' is created and populated with data.

Part 2: Develop OLAP Cube (BIDS)

Step 1: Start BIDS

- Start Menu → SQL Server → SQL Server Business Intelligence Development Studio

Step 2: Create New Project

- File → New → Project → Business Intelligence Projects → Analysis Services Project
- Name it SalesDataAnalysis → Click OK

Step 3: Create Data Source

- Right-click Data Sources → New Data Source → Next → New Connection
- Enter server name, authentication, and select database Sales_DW
- Test Connection → OK → Select Inherit → Give name → Finish

Step 4: Create Data Source View (DSV)

- Right-click Data Source Views → New DSV
- Select Sales_DW → Add FactProductSales table
- Click Add Related Tables → Name it SalesDW_DSV → Finish

Step 5: Create Cube

- Right-click Cubes → New Cube → Use Existing Tables
- Select FactProductSales as Fact Table
- Select Measures and Dimensions → Name: SalesAnalyticalCube → Finish

Step 6: Modify Dimension

- Open Dim Product → Drag Product Name into the Attribute Pane

Step 7: Create Date Hierarchy

- Open Dim Date → Drag in order: Year → Quarter Name → Month Name → Week of Month → Full Date UK

Step 8: Deploy Cube

- Right-click Project → Properties

- Set server name, Deployment Mode: Deploy All, Processing Option: Do Not Process
- Click Deploy → Wait for 'Deployment Completed'

Step 9: Process Cube

- Right-click Project → Process → Click Run
- Wait until status shows: Process Succeeded

Step 10: Browse Cube

- Right-click Cube → Browse
- Drag Product Name → Column, Full Date UK → Row, FactProductSalesCount → Details
- Analysis view is now ready.

End of Practical 9